

BCPD Python Developer

Practice-exam questions and answers

133 questions

Correct options are marked in **green**.

These are the practice-exam questions. Four keys disagreed with the interpreter when the code was run and are corrected here, each one saying what changed. There are no GUI Programming questions in this set, so it does not cover that part of the exam.

Practice-exam set (133)

Domain 1 — Introduction and Setup (37)

1.

Which of the following is not a version of Python?

- A. Python 2
- B. Python 3
- C. Python X**
- D. Python 4

Answer: C

2.

What will the value of the i variable be when the following loop finishes its execution?

```
for i in range(10):  
    pass
```

- A. 10
- B. the variable becomes unavailable
- C. 11
- D. 9**

Answer: D

3.

The following expression 1+-2 is:

- A. equal to 1
- B. invalid
- C. equal to 2
- D. equal to -1**

Answer: D

4.

A compiler is a program designed to

- A. rearrange the source code to make it clearer
- B. check the source code in order to see if it's correct**
- C. execute the source code
- D. translate the source code into machine code**

Answer: B, D

5.

What is the output of the following piece of code?

```
a='ant'  
b='bat'  
c='camel'  
print(a,b,c,sep='')
```

- A. ant' bat' camel
- B. ant'bat'camel
- C. antbatcamel**
- D. SyntaxError

Answer: C

6.

What is the expected output of the following snippet?

```
i=5  
while i>0:  
    i=i//2  
    if i%2==0:  
        break  
    else:  
        i+=1  
print(i)
```

- A. the code is erroneous**
- B. 3
- C. 7
- D. 15

Answer: A

7.

How many lines does the following snippet output?

```
for i in range(1,3):  
    print('**',end='')  
else:  
    print('**')
```

- A. three
- B. one
- C. two**
- D. four

Answer: C

8.

Which of the following literals reflect the value given as 34.23?

- A. 3.42E+01**
- B. 3.42E+01**
- C. 3.42E-03
- D. 3.42E+05

Answer: A, B

9.

Select the valid fun() invocations:

```
def fun(a,b=0):  
    return a*b
```

- A. fun(b=1)
- B. fun(a=0)**
- C. fun(b=1,0)
- D. fun(1)**

Answer: B, D

10.

What is the expected output of the following code?

```
def f(n):  
    if n==1:  
        return '1'  
    return str(n)+f(n-1)  
print(f(2))
```

- A. 21**
- B. 12
- C. 3
- D. 1

Answer: A

11.

What is the expected behaviour of the following snippet?

```
def x():  
    return 2  
x=1+x()  
print(x)
```

- A. cause a runtime exception on line 02
- B. cause a runtime exception on line 01
- C. cause a runtime exception on line 03
- D. print 3**

Answer: D

12.

What is the expected behaviour of the following code?

```
def f(n):  
    for i in range(1,n+1):  
        yield i  
print(f(2))
```

- A. print 4321
- B. print <generator object f at (some hex digits)>**
- C. cause a runtime exception
- D. print 1234

Answer: B

13.

If you need a function that does nothing, what would you use instead of XXX?

```
def idler(): XXX
```

- A. pass
- B. return
- C. exit
- D. None

Answer: A, B

14.

Which of the following words can be used as a variable name? (Select two valid names)

- A. for
- B. TRUE
- C. TRUE
- D. For

Answer: C, D

15.

Python strings can be 'glued' together using the operator:

- A. &
- B. *
- C. ",
- D. +

Answer: D

16.

A keyword

- A. can be used as an identifier
- B. is defined by Python's lexis
- C. is also known as a reserved word
- D. cannot be used in the user's code

Answer: B, C

17.

How many stars (*) does the snippet print?

```
S='*****'  
print(s)
```

- A. the code is erroneous
- B. five
- C. four
- D. None (11 stars)

Answer: A

18.

Assuming that V holds integer 2, which operator should be used instead of OPER to make V OPER 1 equal to 1?

- A. «<
- B. »>
- C. »

D. «

Answer: C

19.

How many stars (*) does the following snippet print?

```
i=3
while i>0:
    i-=1
    print('*')
else:
    print('*')
```

- A. the code is erroneous
- B. five
- C. three
- D. four**

Answer: D

20.

UNICODE is:

- A. the name of an operating system
- B. a standard for encoding and handling texts**
- C. the name of a programming language
- D. the name of a text processor

Answer: B

21.

Which of the equations are True?

- A. chr(ord(x)) == x**
- B. ord(ord(x)) == x
- C. chr(chr(x)) == x
- D. ord(chr(x)) == x**

Answer: A, D

22.

A two-parameter lambda function raising its first parameter to the power of the second parameter should be declared as:

- A. lambda (x,y)=x**y
- B. lambda (x,y): x**y
- C. def lambda (x,y): return x**y
- D. lambda x, y: x**y**

Answer: D

23.

What is the expected output of the following code?

```
def f(n):
    if n==1:
        return 1
    return n+f(n-1)
print(f(2))
```

- A. 21
- B. 12

- C. 3
- D. none

Answer: C

24.

Which function or operator should you use to obtain the answer True or False to the question: 'Do two variables refer to the same object?'

- A. The = operator
- B. The isinstanceOf function
- C. The id() function
- D. The is operator

Answer: D

25.

Select the true statements related to PEP 8 programming recommendations.

- A. You should use the not ... is operator rather than the is not operator
- B. You should make object type comparisons using the isinstance() method
- C. You should write code in a way that favours the CPython implementation over PyPy, Cython, and Jython
- D. You should not write string literals that rely on significant trailing whitespace

Answer: B, D

26.

Which of the following statements are true? (Choose two.)

- A. ASCII stands for Information Interchange
- B. a code point is a number assigned to a given character
- C. ASCII is synonymous with UTF-8
- D. \e is an escape sequence used to mark the end of lines

Answer: A, B

27.

What is the expected output of the following code?

```
myTu=('a','b','c')
m=tuple(map(lambda x: chr(ord(x)+1), myTu))
print(m[-2])
```

- A. a
- B. c
- C. an exception is raised
- D. b

Answer: B

Corrected. The bank keys b. Running it: ('a','b','c') maps through chr(ord(x)+1) to ('b','c','d'), so m[-2] is c. Option B.

28.

Assuming that the following code has been executed successfully, which of the expressions evaluate to True? (Choose two.)

```
def f(x,y):
    nom,denom=x,y
    def g():
        return nom/denom
    return g
a=f(1,2)
b=f(3,4)
```

- A. `b()==4`
- B. `a!=b`**
- C. `a is not None`
- D. `a()==4`

Answer: B, C

29.

`def foo(x,y): return y(x)+(x+1) print(foo(1, lambda x: x*x))`

```
def foo(x,y):  
    return y(x)+(x+1)  
print(foo(1, lambda x: x*x))
```

- A. 3**
- B. 5
- C. 4
- D. an exception is raised

Answer: A

30.

Which of the following lambda definitions are correct? (Choose two.)

- A. `lambda x,y: (x,y)`**
- B. `lambda x,y: return x/y - x%y`
- C. `lambda x,y: x/y - x%y`**
- D. `lambda x,y= x/y - x%y`

Answer: A, C

31.

`x = 3 % 1 y = 1 if x > 0 else 0 print(y)`

```
x = 3 % 1  
y = 1 if x > 0 else 0  
print(y)
```

- A. the code is erroneous and it will not execute
- B. it outputs 1
- C. it outputs -1
- D. it outputs 0**

Answer: D

32.

`x = 8 ** (1/3) y = 2. if x < 2.3 else 3. print(y)`

```
x = 8 ** (1/3)  
y = 2. if x < 2.3 else 3.  
print(y)
```

- A. the code is erroneous and it will not execute
- B. it outputs 2.0**
- C. it outputs 2.5
- D. it outputs 3.0

Answer: B

33.

Which of the following statements are true? (Choose two.)

- A. an escape sequence can be recognised by the / sign put in front of it
- B. ASCII stands for Internal Information
- C. ASCII is a subset of UNICODE
- D. a code point is a number assigned to a given character

Answer: C, D

34.

```
string='123' dummy=0 for character in reversed(string): dummy += int(character) print(dummy)
```

```
string='123'
dummy=0
for character in reversed(string):
    dummy += int(character)
print(dummy)
```

- A. it outputs 321
- B. it outputs 123
- C. it outputs 6
- D. it raises an exception

Answer: C

35.

Which of the following expressions evaluate to True? (Choose two.)

- A. `ord('2') - ord('2') == ord('0')`
- B. `len('6E6E6E6E') > 0`
- C. `chr(ord('a')+1) == 'B'`
- D. `len("") == 1`

Answer: A, B

36.

```
def foo(x,y): return y(x)+y(x+1) print(foo(1, lambda x: x*x))
```

```
def foo(x,y):
    return y(x)+y(x+1)
print(foo(1, lambda x: x*x))
```

- A. 4
- B. 3
- C. an exception is raised
- D. 5

Answer: D

37.

Which of the following expressions evaluate to True? (Choose two.)

- A. `ord('2') - ord('2') == ord('0')`
- B. `chr(ord('A')+1) == 'B'`
- C. `len("") == 1`
- D. `len('0') == ('E'*'E')`

Answer: A, B

Domain 2 — Data Structures (20)

38.

Assuming that the following snippet has been successfully executed, which of the equations are True?

```
a=[1]
b=a
a[0]=0
```

- A. `len(a)==len(b)`
- B. `b[0]+1==a[0]`
- C. `a[0]==b[0]`
- D. `a[0]+1==b[0]`

Answer: A, C

39.

Assuming that the following snippet has been successfully executed, which of the equations are False?

```
a=[0]
b=a[: ]
a[0]=1
```

- A. `len(a)==len(b)`
- B. `a[0]-1==b[0]`
- C. `a[0]==b[0]`
- D. `b[0]-1==a[0]`

Answer: C, D

40.

Which of the following statements are false?

- A. Python strings are actually lists
- B. Python strings can be concatenated
- C. Python strings can be sliced like lists
- D. Python strings are mutable

Answer: A, D

41.

Which of the following sentences are true?

- A. Lists may not be stored inside tuples
- B. Tuples may be stored inside lists
- C. Tuples may not be stored inside tuples
- D. Lists may be stored inside lists

Answer: B, D

42.

Assuming that String is six or more letters long, the following slice `string[1:-2]` is shorter than the original string by:

- A. four chars
- B. three chars
- C. one char
- D. two chars

Answer: B

43.

What is the expected output of the following snippet?

```
lst=[1,2,3,4]
lst=lst[-3:-2]
lst=lst[-1]
print(lst)
```

- A. 1
- B. 4
- C. 2
- D. 3

Answer: C

44.

How many elements will the list2 list contain after execution of the following snippet?

```
list1=[False for i in range(1,10)]
list2=list1[-1:1:-1]
```

- A. zero
- B. five
- C. seven
- D. three

Answer: C

45.

What would you use instead of XXX if you want to check whether a certain key exists in a dictionary called dict?

- A. 'key' in dict
- B. dict['key'] != None
- C. dict.exists('key')
- D. 'key' in dict.keys()

Answer: A, D

46.

You need data which can act as a simple telephone directory. You can obtain it with the following clauses (Select two relevant variants; assume no other items have been created)

- A. dir={'Mom':5551234567,'Dad':5557654321}
- B. dir={'Mom':5551234567,'Dad':5557654321'}
- C. dir={Mom:5551234567,Dad:5557654321}
- D. dir={Mom:'5551234567',Dad:'5557654321'}

Answer: A, B

47.

What is the expected output of the following code?

```
str='abcdef'
def fun(s):
    del s[2]
    return s
print(fun(str))
```

- A. abcef
- B. The program will cause a runtime exception/error
- C. acdef
- D. abdef

Answer: B

48.

Which of the listed actions can be applied to the following tuple?

```
tup=()
```

- A. `tup[0]`
- B. `tup.append(0)`
- C. `tup[0]`
- D. `del tup`

Answer: C, D

49.

Executing the snippet `dct={'pi':3.14}; dct['pi']=3.1415` will cause the `dct`:

- A. to hold two keys named 'pi' linked to 3.14 and 3.1415 respectively
- B. to hold two key named 'pi' linked to 3.14 and 3.1415
- C. to hold one key named 'pi' linked to 3.1415
- D. to hold two keys named 'pi' linked to 3.1415

Answer: C

50.

How many elements will the `list1` list contain after execution of the following snippet?

```
list1 = 'don't think twice, do it!'.split(',')
```

- A. two
- B. zero
- C. one
- D. three

Answer: A

51.

If you want to transform a string into a list of words, what invocation would you use? (Expected output: The, Catcher, in, the Rye,)

```
S='The Catcher in the Rye'  
L=# put proper invocation
```

- A. `s.split()`
- B. `split(s,')`
- C. `s.split('')`
- D. `split(s)`

Answer: A

52.

Assuming that `lst` is a four-element list, is there any difference between these two statements? `del lst` `del lst[:]`

- A. yes, the first line empties the list, the second deletes the list as a whole
- B. yes, the first line deletes the list as a whole, the second just empties the list
- C. no, there is no difference
- D. yes, the first line deletes the list as a whole, the second removes all elements except the first

Answer: B

53.

Which of the following expressions evaluate to True? (Choose two.)

- A. `str(1-1) in '123456789'`
- B. `'dcb' not in 'abcde'[::-1]`
- C. `'phd' in 'alpha'`
- D. `'True' not in 'False'`

Answer: A, D

54.

What is the expected behaviour of the following code?

```
the_list='alpha;beta;gamma'.split(';')
the_string='.'.join(the_list)
print(the_string.isalpha())
```

- A. it raises an exception
- B. it outputs True
- C. it outputs False
- D. it outputs nothing

Answer: C

55.

Assuming that the snippet below has been executed successfully, which of the following expressions evaluate to True? (Choose two.)

```
string='python'[:2]
string=string[-1]+string[-2]
```

- A. `string[0]=='o'`
- B. `string is None`
- C. `len(string)==3`
- D. `string[0]==string[-1]`

Answer: A

56.

Which of the following expressions evaluate to True? (Choose two.)

- A. `'in' in 'Thames'`
- B. `'in' in 'in'`
- C. `'in not' in 'not'`
- D. `'t'.upper() in 'Thames'`

Answer: B, D

57.

Which of the following invocations are valid? (Choose two.)

- A. `sort('python')`
- B. `'python'.find('e')`
- C. `'python'.index('eth')`
- D. `'python'.sorted()`

Answer: B, C

Domain 3 — Object-Oriented Programming (32)

58.

Is it possible to safely check if a class/object has a certain attribute?

- A. yes, by using the hashtart attribute
- B. yes, by using the hashtart() method
- C. yes, by using the hasattr() function**
- D. no, it is not possible

Answer: C

59.

The first parameter of each method:

- A. holds a reference to the currently processed object**
- B. is always set to None
- C. is set to a unique random value
- D. is set by the first argument's value

Answer: A

60.

The simplest possible class definition in Python can be expressed as:

- A. class X:
- B. class X: pass**
- C. class X: return
- D. class X: {}

Answer: B

61.

A variable stored separately in every object is called:

- A. there are no such variables, all variables are shared among objects
- B. a class variable
- C. an object variable
- D. an instance variable**

Answer: D

62.

The following class hierarchy is given. What is the expected output of the code?

```
class A:
    def a(self):
        print('A',end=' ')
    def b(self):
        self.a()
...
B().do()
C().do()
```

- A. BB
- B. CC
- C. AA
- D. BC**

Answer: D

63.

Scenario: You are developing a game in Python and want to represent different characters as objects. Which OOP concept will you use?

- A. Inheritance
- B. Encapsulation
- C. Polymorphism**
- D. Abstraction

Answer: C

64.

If any of a class's components has a name that starts with two underscores (`__`), then:

- A. the class component's name will be mangled**
- B. the class component has to be an instance variable
- C. the class component has to be a class variable
- D. the class component has to be a method

Answer: A

65.

A function called `issubclass(c1,c2)` is able to check if:

- A. c1 and c2 are both subclasses of the same superclass
- B. c2 is a subclass of c1
- C. c1 is a subclass of c2**
- D. c1 and c2 are not subclasses of the same superclass

Answer: C

66.

A class constructor

- A. can return a value
- B. cannot be invoked directly from inside the class**
- C. can be invoked directly from any of the subclasses**
- D. can be invoked directly from any of the superclasses

Answer: B, C

67.

The following class definition is given. We want the `show()` method to invoke the `get()` method, and then output the value the `get()` method returns. Which invocation should be used instead of XXX?

```
class Class:
    def __init__(self, val):
        self.val=val
    def get(self):
        return self.val
    def show(self):
        XXX
```

- A. `print(get(self))`
- B. `print(self.get())`**
- C. `print(get())`
- D. `print(self.get(val))`

Answer: B

68.

What is the expected output of the following snippet?

```
class X: pass
class Y(X): pass
class Z(Y): pass
X=Z()
Z=Z()
print(isinstance(X,Z), isinstance(Z,X))
```

- A. True False
- B. True True
- C. False False
- D. False True**

Answer: D

69.

Which sentence about the @property decorator is false?

- A. The @property decorator should be defined after the method that is responsible for setting an encapsulated attribute.**
- B. The @property decorator designates a method which is responsible for returning an attribute value.
- C. The @property decorator marks the method whose name will be used as the name of the instance attribute.
- D. The @property decorator should be defined before the methods that are responsible for setting and deleting an encapsulated attribute.

Answer: A

70.

Select the true statement about the __name__ attribute.

- A. __name__ is a special attribute, inherent for both classes and instances, and it contains information about the class to which a class instance belongs.
- B. __name__ is a special attribute, inherent for both classes and instances, and it contains a dictionary of object attributes.
- C. __name__ is a special attribute, inherent for classes and it contains information about the class to which a class instance belongs.
- D. __name__ is a special attribute, inherent for classes, and it contains the name of a class.**

Answer: D

71.

What is the expected behaviour of the following code? (missing colon after Sub_B(Super))

- A. it outputs 0
- B. it outputs 1
- C. it raises an exception**
- D. it outputs 2

Answer: C

72.

Assuming that the following inheritance set is in force, which of the following classes are declared properly? (Choose two.)

```
class A: pass
class B(A): pass
class C(A): pass
class D(B): pass
```

- A. class Class_4(D,A): pass**

- B. class Class_3(A,C): pass
- C. class Class_2(B,D): pass**
- D. class Class_1(C,D): pass

Answer: A, C

73.

```
class Upper: def method(self): return 'upper' class Lower(Upper): def method(self): return 'lower' Object=Upper() print(isinstance(Object, Lower), end=' ') print(Object.method())
```

```
class Upper:
    def method(self):
        return 'upper'
class Lower(Upper):
    def method(self):
        return 'lower'
Object=Upper()
print(isinstance(Object,Lower), end=' ')
print(Object.method())
```

- A. True upper
- B. True lower
- C. False upper**
- D. False lower

Answer: C

74.

What is the expected behaviour of the following code? (o1.foo missing parentheses)

- A. it outputs 1
- B. it outputs 3
- C. it outputs 6
- D. it raises an exception**

Answer: D

75.

```
class Class: Variable=0 def __init__(self): self.value=0 object_1=Class() object_1.Variable+=1 object_2=Class() object_2.value+=1 print(object_2.Variable+object_1.value)
```

```
class Class:
    Variable=0
    def __init__(self):
        self.value=0
object_1=Class()
object_1.Variable+=1
object_2=Class()
object_2.value+=1
print(object_2.Variable+object_1.value)
```

- A. it outputs 0**
- B. it raises an exception
- C. it outputs 1
- D. it outputs 2

Answer: A

76.

What is true about Object-Oriented Programming in Python? (Choose two.)

- A. each object of the same class can have a different set of methods
- B. a subclass is usually more specialised than its superclass**
- C. if a real-life object can be described with a set of adjectives, they may reflect a Python object method
- D. the same class can be used many times to build a number of objects**

Answer: B, D

77.

What is true about Python class constructors? (Choose two.)

- A. there can be more than one constructor in a Python class
- B. the constructor must return a value other than None
- C. the constructor is a method named __init__**
- D. the constructor must have at least one parameter**

Answer: C, D

78.

Assuming that the following piece of code has been executed successfully, which of the expressions evaluate to True? (Choose two.)

```
class A:
    VarA=1
    def __init__(self):
        self.prop_a=1
class B(A):
    VarA=2
    def __init__(self):
        super().__init__()
        self.prop_b=2
obj_a=A()
obj_aa=A()
obj_b=B()
obj_bb=obj_b
```

- A. isinstance(obj_b,A)**
- B. A.VarA==1**
- C. obj_a is obj_aa
- D. B.VarA==1

Answer: A, B

79.

class Class: Var=data=1 def __init__(self, value): self.prop=value Object=Class(2) Which are true? (Choose two.)

```
class Class:
    Var=data=1
    def __init__(self, value):
        self.prop=value
Object=Class(2)
Which are true? (Choose two.)
```

- A. len(Class.__dict__)==1
- B. 'data' in Class.__dict__**
- C. 'var' in Class.__dict__
- D. 'data' in Object.__dict__

Answer: B

80.

A property that stores information about a given class's super-classes is named:

- A. `__upper__`
- B. `__super__`
- C. `__ancestors__`
- D. `__bases__`

Answer: D

81.

```
class Class: Var=0 def _foo(self): Class.Var+=1 return Class.Var o=Class() o._Class_foo() print(o._Class_foo())
```

```
class Class:
    Var=0
    def _foo(self):
        Class.Var+=1
        return Class.Var
o=Class()
o._Class_foo()
print(o._Class_foo())
```

- A. it raises an exception
- B. it outputs 3
- C. it outputs 1
- D. it outputs 6

Answer: A

82.

The `__bases__` property contains:

- A. base class location (addr)
- B. base class objects (class)
- C. base class names (str)
- D. base class ids (int)

Answer: B

83.

```
class Super: def make(self): pass def doit(self): return self.make() class Sub_A(Super): def make(self): return 1 class Sub_B(Super): pass a=Sub_A() b=Sub_B() print(a.doit()+b.doit())
```

```
class Super:
    def make(self): pass
    def doit(self):
        return self.make()
class Sub_A(Super):
    def make(self):
        return 1
class Sub_B(Super):
    pass
a=Sub_A()
b=Sub_B()
print(a.doit()+b.doit())
```

- A. It outputs 0
- B. It raises an exception
- C. It outputs 1
- D. It outputs 2

Answer: B

84.

Assuming that the code below has been executed successfully, which of the expressions evaluate to True? (Choose two.)

```
class Class:
    var=1
    def __init__(self, value):
        self.prop=value
Object=Class(2)
```

- A. 'var' in Class.__dict__
- B. 'var' in Object.__dict__
- C. len(Object.__dict__) == 1
- D. 'prop' in Class.__dict__

Answer: A, C

85.

What is true about Python class constructors? (Choose two.)

- A. there can be only one constructor in a Python class
- B. the constructor cannot be invoked directly under any circumstances
- C. the constructor cannot return a result other than None
- D. the constructor's first parameter must always be named self

Answer: B, D

86.

Assuming that the following inheritance set is in force, which of the following classes are declared properly? (Choose two.)

```
class A: pass
class B(A): pass
class C(A): pass
class D(B,C): pass
```

- A. class Class_3(A,C): pass
- B. class Class_4(C,B): pass
- C. class Class_1(D): pass
- D. class Class_2(A,B): pass

Answer: A, C

87.

```
class Class: Variable=0
def __init__(self):
    self.value=0
object_1=Class()
Class.Variable+=1
object_2=Class()
object_2.value+=1
print(object_2.Variable+object_1.value)
```

```
class Class:
    Variable=0
    def __init__(self):
        self.value=0
object_1=Class()
Class.Variable+=1
object_2=Class()
object_2.value+=1
print(object_2.Variable+object_1.value)
```

- A. It outputs 2
- B. It raises an exception
- C. It outputs 1
- D. It outputs 0

Answer: C

88.

```
class Upper:
    def __init__(self):
        self.property='upper'
class Lower(Upper):
    def __init__(self):
        super().__init__()
Object=Lower()
print(isinstance(Object,Lower), end="")
print(Object.property)
```

```
class Upper:
    def __init__(self):
        self.property='upper'
class Lower(Upper):
    def __init__(self):
        super().__init__()
Object=Lower()
print(isinstance(Object,Lower), end='')
print(Object.property)
```

- A. True lower
- B. True upper**
- C. False upper
- D. False lower

Answer: B

89.

```
class ClassA:
    var=1
    def __init__(self,prop):
        prop1=prop2=prop
class ClassB(ClassA):
    def __init__(self,prop):
        prop3=prop**2
        super().__init__(prop)
    def __str__(self):
        return 'Object'
Object=ClassA(2)
```

```
class ClassA:
    var=1
    def __init__(self,prop):
        prop1=prop2=prop
class ClassB(ClassA):
    def __init__(self,prop):
        prop3=prop**2
        super().__init__(prop)
    def __str__(self):
        return 'Object'
Object=ClassA(2)
```

- A. ClassA.__module__ == '__main__'
- B. __name__ == '__main__'
- C. str(Object) == 'Object'**
- D. len(ClassB.__bases__) == 2

Answer: A, C

Domain 4 — Modules and Libraries (17)

90.

Can a module run like regular code?

- A. yes, and it can differentiate its behaviour between the regular launch and import**
- B. it depends on the Python version
- C. yes, but it cannot differentiate its behaviour between the regular launch and import
- D. no, it is not possible; a module can be imported, not run

Answer: A

91.

A file name like services.cpython-36.pyc says that:

- A. the interpreter used to generate the file is version 3.6**
- B. it has been produced by CPython**

- C. it is the 36th version of the file
- D. the file comes from the services.py source file**

Answer: A, B, D

92.

What can you do if you don't like a long package path like `import alpha.beta.gamma.delta.epsilon.zeta`?

- A. you can make an alias for the name using the `alias` keyword
- B. nothing, you need to come to terms with it
- C. you can shorten it to `alpha.zeta` and Python will find the proper connection
- D. you can make an alias for the name using the `as` keyword**

Answer: D

93.

What should you put instead of XXX to print out the module name?

```
| if __name__ != 'XXX':  
|     print(__name__)
```

- A. `main`
- B. `_main`
- C. `__main__`**
- D. `_main_`

Answer: C

94.

Files with the suffix `.py` contain:

- A. Python source code**
- B. backups
- C. temporary data
- D. semi-compiled Python code

Answer: A

95.

Package source directories/folders can be:

- A. converted into the so-called pycK format
- B. packed as a ZIP file and distributed as one file**
- C. rebuilt to a flat form and distributed as one directory/folder
- D. removed as Python compiles them into an internal portable format

Answer: B

96.

What can you deduce from the line below?

```
| x = a.b.c.f()
```

- A. `import a.b.c` should be placed before that line**
- B. `f()` is located in subpackage `c` of subpackage `b` of package `a`**
- C. the line is incorrect
- D. the function being invoked is called `a.b.c.f()`**

Answer: A, B, D

97.

Assuming that the code below has been executed successfully, which of the following expressions will always evaluate to True? (Choose two.)

```
import random
random.seed(1)
v1=random.random()
random.seed(1)
v2=random.random()
```

- A. `v1 == v2`
- B. `v1 > v2`
- C. `len(random.sample([1,2,3],2)) > 2`
- D. `random.choice([1,2,3]) >= 1`

Answer: A, D

98.

Which one of the platform module functions should be used to determine the underlying platform name?

- A. `platform.python_version()`
- B. `platform.processor()`
- C. `platform.platform()`
- D. `platform.uname()`

Answer: C

99.

What is the expected output of the following code?

```
import sys
import math
b1=type(dir(math)[0]) is str
b2=type(sys.path[-1]) is str
print(b1 and b2)
```

- A. FALSE
- B. None
- C. TRUE
- D. 0

Answer: C

100.

With regards to the directory structure, select the proper forms of the directives in order to import module_a. (Choose two.)

- A. `from pypack import module_a`
- B. `import module_a from pypack`
- C. `import module_a`
- D. `import pypack.module_a`

Answer: A, D

101.

A Python module named pymod.py contains a function named python(). Which of the following snippets will let you invoke the function? (Choose two.)

- A. `import pymod; pymod.python()`
- B. `from pymod import python; python()`
- C. `from pymod import *; pymod.python()`
- D. `import python from pymod; python()`

Answer: A, B

102.

What is true about Python packages? (Choose two.)

- A. a package is a single file whose name ends with the .pa extension
- B. a package is a group of related modules**
- C. the __name__ variable always contains the name of a package**
- D. the .pyc extension is used to mark semi-compiled Python packages

Answer: B, C

103.

A Python module named pymod.py contains a variable named pymod. Which of the following snippets will let you access the variable? (Choose two.)

- A. import pymod; pymod.pyvar = 1
- B. import pymod from pymod; pymod = 1
- C. from pymod import pymod; pymod()**
- D. from pymod import pymod; pymod = 1**

Answer: C, D

104.

Assuming that the code below has been executed successfully, which of the following expressions will always evaluate to True? (Choose two.)

```
import random
v1=random.random()
v2=random.random()
```

- A. v1 == v2
- B. v1 < 1**
- C. random.choice([1,2,3]) > 0**
- D. len(random.sample([1,2,3],1)) > 2

Answer: B, C

105.

With regards to the directory structure, select the proper forms of the directives in order to import module_b. (Choose two.)

- A. from pyback.upper import module_b
- B. import pyback.upper.module_b**
- C. import upper.module_b**
- D. import module_b

Answer: B

106.

import sys b1=type(dir(sys)) is str b2=type(sys.path[-1]) is str print(b1 and b2)

```
import sys
b1=type(dir(sys)) is str
b2=type(sys.path[-1]) is str
print(b1 and b2)
```

- A. FALSE**
- B. 0
- C. None
- D. TRUE

Answer: A

The bank's key for this one reads *FALSE* — the option **text** rather than a letter, and almost certainly an Excel coercion of "False" into a boolean. It maps to option A. Running it confirms: `dir(sys)` returns a list, not a str, so `b1` is False and `b1 and b2` is *False*.

Domain 5 — Debugging and Error Handling (16)

107.

What is the expected output of the following snippet?

```
s='abc'
for i in len(s):
    s[i]=s[i].upper()
print(s)
```

- A. abc
- B. The code will cause a runtime exception**
- C. ABC
- D. 123

Answer: B

108.

What is the expected behaviour of the following snippet?

```
def a(l,l):
    return l[l]
print(a(0,[1]))
```

- A. cause a runtime exception
- B. print 1
- C. print 0, [1]
- D. SyntaxError**

Answer: D

109.

How to handle an exception and assign it to variable e?

- A. `except Exception (e):`
- B. `except e = Exception:`
- C. `except Exception as e:`**
- D. such an action is not possible in Python

Answer: C

110.

What is the expected output of the following code?

```
lst=[x for x in range(5)]
lst=list(filter(lambda x: x%2==0, lst))
print(len(lst))
```

- A. 2**
- B. The code will cause a runtime exception
- C. 1
- D. SyntaxError

Answer: A

111.

What is the expected behaviour of the following code?

```
def unclear(x):  
    if x%2==1:  
        return 0  
print(unclear(1)+unclear(2))
```

- A. print 0
- B. cause a runtime exception**
- C. prints 3
- D. print an empty line

Answer: B

112.

If you need to serve two different exceptions called Ex1 and Ex2 in one except branch, you can write:

- A. except Ex1 Ex2:
- B. except (Ex1, Ex2):**
- C. except Ex1, Ex2:
- D. except Ex1+Ex2:

Answer: B

113.

What is the expected behaviour of the following code? (try/except missing colon)

- A. it outputs 3
- B. it outputs 1
- C. the code is erroneous and it will not execute**
- D. it outputs 2

Answer: C

114.

What is the expected behaviour of the following code?

```
string=str(1/3)  
dummy='',  
for character in strong:  
    dummy=character+dummy  
print(dummy[-1])
```

- A. it raises an exception**
- B. it outputs 0
- C. it outputs 3
- D. it outputs None

Answer: A

115.

`my_list=[i for i in range(5)] m=[my_list[i] for i in range(4,0,-1)] if my_list[i]%2!=0] print(m)`

```
my_list=[i for i in range(5)]  
m=[my_list[i] for i in range(4,0,-1)] if my_list[i]%2!=0]  
print(m)
```

- A. it outputs [1,3]
- B. the code is erroneous and it will not execute**
- C. it outputs [3,1]
- D. it outputs [4,2,0]

Answer: B

116.

Which of the following snippets will execute without raising any unhandled exceptions? (Choose two.)

- A. `try: print(float('1e1')) except (ValueError,NameError): print(float('1a1')) else: print(float('101'))`
- B. `try: print(1/1) except: print(2/1) else: print(3/0)`
- C. `try: print(1/0) except ValueError: print(1/1) else: print(1/2)`
- D. `try: print(0/1) except: print(1/1) else: print(2/1)`

Answer: A, D

117.

`my_list=[i for i in range(5,0,-1)] m=[my_list[i] for i in range(5)] if my_list[i]%2==0] print(m)`

```
my_list=[i for i in range(5,0,-1)]
m=[my_list[i] for i in range(5)] if my_list[i]%2==0]
print(m)
```

- A. it outputs [4,2]
- B. it outputs [2,4]
- C. it outputs [0,1,2,3,4]
- D. the code is erroneous and it will not execute

Answer: D

118.

`the_list='1,2 3'.split() the_string=''.join(the_list) print(the_string.insight())`

```
the_list='1,2 3'.split()
the_string=''.join(the_list)
print(the_string.insight())
```

- A. it raises an exception
- B. it outputs nothing
- C. it outputs True
- D. it outputs False

Answer: A

119.

`s='2A' try: n=int(s) except ValueError: n=2 except ArithmeticError: n=1 except: n=0 print(n)`

```
s='2A'
try:
    n=int(s)
except ValueError:
    n=2
except ArithmeticError:
    n=1
except:
    n=0
print(n)
```

- A. it outputs 0
- B. the code is erroneous and it will not execute
- C. it outputs 1
- D. it outputs 2

Answer: D

120.

Which of the following snippets will execute without raising any unhandled exceptions? (Choose two.)

- A. `try: print(0/0) except: print(0/1) else: print(0/2)`
- B. `try: print(int('0')) except NameError: print('0') else: print(int(''))`
- C. `import math try: print(math.sqrt(-1)) except: print(math.sqrt(0)) else: print(math.sqrt(1))`
- D. `try: print(float('1e1')) except (NameError, SystemError): print(float('1a1')) else: print(float('1c1'))`

Answer: A, C

121.

`my_list=[1,2,3] try: my_list[3]=my_list[2] except BaseException as error: print(error)`

```
my_list=[1,2,3]
try:
    my_list[3]=my_list[2]
except BaseException as error:
    print(error)
```

- A. it outputs error
- B. it outputs `<class 'IndexError'>`
- C. it outputs list assignment index out of range
- D. the code is erroneous and it will not execute

Answer: C

122.

`class E(Exception): def __init__(self,message): self.message=message def __str__(self): return 'it's nice to see you' try: print('I feel fine') raise Exception('what a pity') except E as e: print(e) else: print('the show must go on')`

```
class E(Exception):
    def __init__(self,message):
        self.message=message
    def __str__(self):
        return 'it's nice to see you'
try:
    print('I feel fine')
    raise Exception('what a pity')
except E as e:
    print(e)
else:
    print('the show must go on')
```

- A. the string 'it's nice to see you' will be seen
- B. the string 'the show must go on' will be printed
- C. the string 'I feel fine' will be printed
- D. nothing will be printed

Answer: C

Domain 6 — File Handling (11)

123.

There is a stream named `s` open for writing. What option will you select to write a line to the stream?

- A. `s.write('Hello\n')`
- B. `write(s,'Hello')`
- C. `s.writeln('Hello')`
- D. `s.writeln('Hello')`

Answer: A

124.

You are going to read just one character from a stream called `s`. Which statement would you use?

- A. `ch = read(s,1)`
- B. `ch = s.input(1)`
- C. `ch = input(s,1)`
- D. `ch = s.read(1)`**

Answer: D

125.

What can you deduce from the following statement?

```
| str = open('file.txt', 'rt')
```

- A. `str` is a string read in from the file named `file.txt`
- B. a newline character translation will be performed during the reads**
- C. if `file.txt` does not exist, it will be created
- D. the opened file cannot be written with the use of the `str` variable**

Answer: B, D

126.

What does the `open()` function return?

- A. an integer value identifying an opened file
- B. an error code (0 means success)
- C. a stream object**
- D. always `None`

Answer: C

127.

If `S` is a stream open for reading, what do you expect from the following invocation?

```
| c = s.read()
```

- A. one line of the file will be read and stored in the string called `c`
- B. the whole file content will be read and stored in the string called `c`**
- C. one character will be read and stored in the string called `c`
- D. one disk sector (512 bytes) will be read and stored in the string called `c`

Answer: B

128.

You are going to read 16 bytes from a binary file into a bytearray called `data`. Which lines would you use?

- A. `data=bytearray(16); bf.readinto(data)`**
- B. `data=binfile.read(bytearray(16))`
- C. `bf.readinto(data=bytearray(16))`
- D. `data=bytearray(binfile.read(16))`**

Answer: A, D

129.

Story: Sarah is writing a Python program that reads data from a file. Which of the following modes should she use to open the file for reading?

- A. `'w'`
- B. `'r'`**
- C. `'a'`

D. 'x'

Answer: B

130.

```
try: f=open('non_existing_file','w') print(1,end=' ') s=f.readline() print(2,end=' ') except IOError as error: print(3,end=' ')
else: f.close() print(4,end=' ')
```

```
try:
    f=open('non_existing_file','w')
    print(1,end=' ')
    s=f.readline()
    print(2,end=' ')
except IOError as error:
    print(3,end=' ')
else:
    f.close()
    print(4,end=' ')
```

- A. 124
- B. 1234
- C. 24
- D. 13**

Answer: D

Corrected. The bank keys 124. Running it prints 1 3. Opening in 'w' mode succeeds and creates the file, so print(1) runs; then readline() on a write-only handle raises io.UnsupportedOperation, which subclasses OSError and so IS caught by except IOError, printing 3. The else branch never runs because an exception was handled.

131.

```
try: f=open('existing_text_file','w') d=f.readlines() print(len(d)) f.close() except IOError: print(-1)
```

```
try:
    f=open('existing_text_file','w')
    d=f.readlines()
    print(len(d))
    f.close()
except IOError:
    print(-1)
```

- A. the length of the first line from the file
- B. -1**
- C. the number of lines contained inside the file
- D. the length of the last line from the file

Answer: B

The bank's key for this one reads 0 (none of the above), which is not one of the four options. Running it prints -1: 'w' mode truncates the file, readlines() on a write-only handle raises io.UnsupportedOperation (a subclass of OSError), the handler catches it and prints -1. Option B.

132.

Which of the following statements are true? (Choose two.)

- A. if invoking open() fails, an exception is raised**
- B. open() requires a second argument
- C. open() is a function which returns an object that represents a physical file**
- D. instd, outstd, errstd are the names of pre-opened streams

Answer: A, C

133.

```
try: f=open('non_zero_length_existing_text_file','rt') d=f.read(1) print(len(d)) f.close() except IOError: print(-1)
```

```
try:
    f=open('non_zero_length_existing_text_file','rt')
    d=f.read(1)
    print(len(d))
    f.close()
except IOError:
    print(-1)
```

- A. 1
- B. 0
- C. an error value corresponding to file not found
- D. 1

Answer: A